

Neuro-Symbolic Legal Invariant Checker

Complete Plain-English Project Guide: Story, Dictionary, Pipeline & Professor Viva Prep

1. The Entire Project in ONE Simple Sentence

"This project is an automated bug-checker for multi-million dollar bank loan contracts — it finds mathematical contradictions that human lawyers accidentally wrote."

Think of it like Grammarly, but instead of checking spelling mistakes, it checks whether the contract's financial math rules clash with each other.

2. Plain-English Dictionary: Financial Words

Word	What It Actually Means	Real-Life Example
Contract / Agreement	A legal deal signed between a Bank and a Company.	The bank says: 'Here is \$50M, but follow our rules.'
Borrower vs. Lender	Borrower = Company taking money. Lender = Bank giving money.	If you take a loan from SBI, you are Borrower, SBI is Lender.
Covenant	A strict promise / rule in the contract.	'Rule 1: Keep your total debt under a safe ceiling.'
EBITDA	Pure Profit from running the business. (Earnings Before Interest, Taxes, Depreciation)	If you sell \$100 of burgers and spent \$60 on ingredients, your EBITDA is \$40.
Total Debt	Total borrowed loan money owed to banks.	Loans from Bank A (\$10M) + Bank B (\$20M) = \$30M Total Debt.
Leverage Ratio	How many times bigger your debt is compared to profit (Debt / EBITDA).	Debt of \$30M divided by Profit of \$10M = 3.0x Leverage Ratio.
Interest Expense	The fee/interest you pay the bank for borrowing.	Paying 5% annual interest on your loan.
Grace Period	Extra days given to pay if you miss a due date.	'15-day cure window before the bank can take legal action.'
Acceleration	The bank demanding ALL the loan money back TODAY.	Instead of monthly installments, the bank says: 'Pay all \$50M right now!'
Redline	The corrected replacement clause written to fix the bug.	Striking through bad contract text and inserting the fix.

3. Plain-English Dictionary: AI & Computer Science Words

Word	What It Actually Means	Real-Life Example
Invariant	A golden safety rule that must NEVER, EVER be broken.	In a car: 'Speed must never exceed 200 km/h.' In a loan: 'Debt must never exceed the cap.'
SMT Solver (Z3)	A supercomputer math prover built by Microsoft Research.	Checks all possible numbers from -inf to +inf to prove if a formula can break.
SAT (Satisfiable)	'I found a bug!' The solver found numbers that break the contract.	'When Profit = \$7.8M, Rule A allows \$31.2M, but Rule B caps at \$30M. SAT!'
UNSAT (Unsatisfiable)	'100% Safe!' It is mathematically impossible to break the rule.	The solver proves zero loopholes exist under any scenario.
Counterexample	The exact mathematical numbers that trigger the loophole.	Proof values: [Total Debt = \$31.2M, EBITDA = \$7.8M].
Neuro-Symbolic	Combining a Reader (Neural) with a Calculator (Symbolic).	Gemini reads English prose -> Z3 solves pure math equations.

AST / DSL	Turning messy English prose into clean mathematical formulas.	Translating 'Debt cannot exceed twice profit' into debt $\leq 2 * \text{profit}$.
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4. The Real-Life Story: Where Does the Bug Come From?

A 100-page loan contract is written by multiple lawyers across different teams. Contradictions happen naturally:

- **Lawyer 1 (Section 6.01, Page 10):** 'During an acquisition, you can borrow up to 4.0 times your profit.'
- **Lawyer 2 (Section 6.02, Page 60):** 'Under no circumstances can total debt exceed \$30,000,000 if profit is under \$8,000,000.'

What happens in real life if the company makes \$7.8 Million in profit?

1. Lawyer 1's rule says the company can legally borrow: $4.0 * \$7.8\text{M} = \text{\$31.2 Million}$.
2. Lawyer 2's rule says debt is strictly capped at: **\\$30.0 Million**.

The company borrows \$31M thinking they followed the rules, but the bank claims they breached the contract! This leads to lawsuits worth millions of dollars. Our software catches this in 19 milliseconds.

5. The Pipeline: How Data Moves From Start to Finish (Step-by-Step)

Here is the complete journey of how a contract moves through your system, explained simply:

Stage	What Happens (In Plain English)	Code Component
Stage 1: Document Ingestion	The user selects or uploads a contract (PDF or TXT). The system reads the pages and extracts the raw legal text into memory.	core/parser.py Uses PyPDF
Stage 2: Neural Translation (Gemini)	Gemini acts as an English-to-Math translator. It ignores fluff words and translates sentences into strict math equations (e.g. 'total_debt <= 4.0 * ebitda').	core/neural_extractor.py Gemini 2.5 Flash
Stage 3: The Intermediate Blueprint (AST)	The equations are stored in clean structured data models: Real numbers for money, Integers for days, and Booleans for yes/no states.	core/dsl_schema.py Pydantic Models
Stage 4: SMT Solving (Z3 Engine)	Microsoft Z3 takes the equations and tests all possible numbers from -inf to +inf. In 19ms, it mathematically proves whether any combination can break the rules.	core/symbolic_verifier.py Z3 Theorem Prover
Stage 5: Legal Diagnosis & Redline	If Z3 finds a bug, it outputs the proof numbers. The explainer converts those numbers into plain English and writes a replacement contract clause to fix it.	core/explainer.py Automated Redliner
Stage 6: Interactive Dashboard	The user sees green/red status badges, solver execution times, metrics, and suggested amendments on a clean web page. Includes an offline mode for zero-internet college demos.	app.py Streamlit Web UI

6. Professor Defense / Viva Voce Cheat Sheet

Q1: What is your project about?

"Sir, financial loan contracts are hundreds of pages long and frequently contain contradictory borrowing limits. My project uses Gemini AI to read the contract into math formulas, and the Z3 SMT solver to mathematically prove whether any loopholes exist with 100% certainty."

Q2: What is an Invariant?

"An invariant is a safety rule that must never be broken under any circumstances — for example, ensuring that a borrower's total debt never exceeds the bank's maximum safety ceiling."

Q3: What is EBITDA?

"EBITDA stands for Earnings Before Interest, Taxes, Depreciation, and Amortization. In simple words, it is the company's operating cash profit from running its core business."

Q4: Why did you use Z3 instead of just asking Gemini?

"Because Gemini is an LLM — it hallucinates and cannot perform mathematical proofs. Z3 is an industrial SMT theorem prover from Microsoft that provides 100% mathematical certainty with zero hallucinations."

Q5: What did YOU do as a researcher?

"I designed the formal domain schema that translates legal English into SMT mathematics, formulated the invariant proof equations, built the Z3 verification engine, and benchmarked that our Neuro-Symbolic approach achieves 100% mathematical soundness compared to 62% for pure LLMs."